SIDE 1

STAT 251 · EXAM 2 — Calculation Sheet

A Which procedure? | 1–4 Module 4 | 5–6 z / t tables | 7 SE · Welch (side 2) | 8, 10 CI (side 2) | 11–13 Tests (side 2) | 14 Worked (side 2) | 15–16 MC (side 2)

A · If the question says…	Parameter → estimate	Two-sided CI (estimate ± crit × SE)	Test statistic (under H_0)	Table · df (conditions → §13)
"% / proportion of all…", one yes/no group	$p \to \hat{p} = \frac{x}{n}$	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$ ← uses p_0!	**z**
compare % of two separate groups	$p_1 - p_2 \to \hat{p}_1 - \hat{p}_2$	$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$	$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$ (lecture: unpooled)	**z**
"average / mean of all…", one group, s given	$\mu \to \bar{y}\ (= \bar{x})$	$\bar{y} \pm t^*_{n-1} \frac{s}{\sqrt{n}}$	$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$	**t**, $df = n - 1$
compare means of two separate groups	$\mu_1 - \mu_2 \to \bar{y}_1 - \bar{y}_2$	$(\bar{y}_1 - \bar{y}_2) \pm t^*_\nu \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$t = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$, Δ_0 usually 0	**t**, Welch ν (§7, side 2) — round **down**
same subjects before/after, matched pairs	$\mu_d \to \bar{d}$ (one list d_i)	$\bar{d} \pm t^*_{n-1} \frac{s_d}{\sqrt{n}}$, n = # pairs	$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}$, d_0 usually 0	**t**, $df = n - 1$
Normal X: P(value…) with μ, σ known	—	reverse: $x = \mu + z\sigma$	$z = \frac{x - \mu}{\sigma}$	**z**
P(sample mean …) with μ, σ known	—	$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$	$z = \frac{\bar{y} - \mu}{\sigma/\sqrt{n}}$	**z**
"how many to sample for margin E?"	proportion	$n = \left\lceil \frac{(z^*)^2 \hat{p}(1-\hat{p})}{E^2} \right\rceil$ — no prior estimate → $\hat{p} = .5$; **always round UP**; E as a decimal (3% = .03)		**z***

POPULATION (parameters & claims)

$\boldsymbol{p}$ = true population proportion (unknown). *"the true % / proportion of all…"*
$\boldsymbol{\mu}$ = true population mean (unknown). *"true average of all…", "mean of the population"*
$\boldsymbol{\sigma}$ = population SD; $\boldsymbol{\sigma^2}$ = variance. *"population SD is…", "SD = 0.04 s"; known $\sigma \to z$*
$\boldsymbol{p_0}$, $\boldsymbol{\mu_0}$ = value claimed in $\boldsymbol{H_0}$ (null value). *"claims 52%", "is the mean 0.400?"*
$\boldsymbol{q_0 = 1 - p_0}$ = complement of the null proportion.
$\boldsymbol{p_1 - p_2}$, $\boldsymbol{\mu_1 - \mu_2}$ = difference between groups 1 and 2. *"compare A vs B"*
$\boldsymbol{\mu_d}$ = true mean of paired differences. *"before/after on the same people"*
$\boldsymbol{\Delta_0}$, $\boldsymbol{d_0}$ = claimed difference in $\boldsymbol{H_0}$ (usually 0). *"no difference"*
$\boldsymbol{N}$ = population size (10% condition).

SAMPLE (what the data give you)

$\boldsymbol{n}$ = sample size (# in sample; paired: # of pairs). *"randomly sampled 36…"*
$\boldsymbol{x}$ = # of "successes" (yes answers). *"62 of 100 prefer…"*
$\boldsymbol{\hat{p} = \frac{x}{n}}$ = sample proportion ("p-hat"). *"62% of those sampled…"*
$\boldsymbol{\hat{q} = 1 - \hat{p}}$ = sample proportion of "no".
$\boldsymbol{\bar{y}\ (= \bar{x})}$ = sample mean ("y-bar"). *"the sample averaged 0.424 s"*
$\boldsymbol{s}$ = sample SD; $\boldsymbol{s^2}$ = sample variance. *"sample standard deviation 0.060" → t*
$\boldsymbol{n_1, \bar{y}_1, s_1, \hat{p}_1}$ = values for group 1 (subscript 2 = group 2).
$\boldsymbol{d_i}$ = one pair's difference (e.g. after − before).
$\boldsymbol{\bar{d}}$, $\boldsymbol{s_d}$ = mean and SD of the $\boldsymbol{d_i}$ list.

INFERENCE TOOLS

PE = point estimate (best single guess: $\hat{p}$, $\bar{y}$, …).
$\boldsymbol{SE}$ = standard error = SD of the estimate.
$\boldsymbol{z^*}$, $\boldsymbol{t^*_{df}}$ = critical value from the confidence level. *"95% confident", "90% CI"*
$\boldsymbol{E}$ (ME, $\boldsymbol{m}$) = margin of error = $\boldsymbol{c^* \times SE}$. *"within ±3 points", "accurate to .01"*
$\boldsymbol{C}$ = confidence level; $\boldsymbol{\alpha = 1 - C}$.
$\boldsymbol{\alpha}$ = significance level (P of Type I error). *"at the 5% level", "α = .05"*
$\boldsymbol{df}$, $\boldsymbol{\nu}$ = degrees of freedom (row of the t-table).
$\boldsymbol{z}$, $\boldsymbol{t}$ = test statistic (# of SEs from the null).
p-value = P(result this extreme or more | $\boldsymbol{H_0}$ true).
$\boldsymbol{H_0}$, $\boldsymbol{H_a}$ = null (has =) / alternative (<, > , ≠). *"test whether…", "is there evidence…"*
$\boldsymbol{\beta}$ = P(Type II); power = $\boldsymbol{1 - \beta}$.

MODULE 4 SYMBOLS

$\boldsymbol{X}$ = the random variable (the quantity measured). *"let X = time until…"*
$\boldsymbol{x}$ = one specific value of $\boldsymbol{X}$.
$\boldsymbol{f(x)}$ = PDF (density curve; area = probability). *"the pdf is…"*
$\boldsymbol{F(x)}$ = CDF = $\boldsymbol{P(X \le x)}$, area to the left. *"the cdf", "at most x"*
$\boldsymbol{E[X]}$, $\boldsymbol{\mathrm{Var}(X)}$ = mean / variance of $\boldsymbol{X}$. *"expected value", "average"*
$\boldsymbol{a}$, $\boldsymbol{b}$ = Uniform min and max. *"anywhere between a and b"*
$\boldsymbol{c}$, $\boldsymbol{d}$ = endpoints of the interval asked about.
$\boldsymbol{\lambda}$ = rate (events per time unit). *"5 per hour", "on average 2 per hour"*
$\boldsymbol{k}$ = unknown constant in a PDF. *"find k so f is a valid pdf"*
$\boldsymbol{Z}$, $\boldsymbol{\Phi(z)}$ = standard Normal; table area left of $\boldsymbol{z}$.
$\boldsymbol{x_p}$, $\boldsymbol{m}$ = $\boldsymbol{p}$-th percentile; median ($\boldsymbol{F(m) = .5}$). *"90th percentile", "median"*

1 Continuous RVs: PDF / CDF toolkit

$$f(x) \ge 0 \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P(X = x) = 0$$

$$P(a < X < b) = \int_a^b f(x)\,dx = F(b) - F(a)$$

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(t)\,dt \qquad f(x) = F'(x)$$

$P(X > x) = 1 - F(x)$ $\quad <$ and $\le$ give the **same** answer

$$\mu = E[X] = \int x\,f(x)\,dx \qquad E[X^2] = \int x^2 f(x)\,dx$$

$$\sigma^2 = \mathrm{Var}(X) = E[X^2] - \mu^2 = \int (x-\mu)^2 f(x)\,dx, \qquad \sigma = \sqrt{\sigma^2}$$

Typical asks → do this:

- **find** k: set $\int_{\text{support}} f(x)\,dx = 1$, solve.
- **CDF from PDF**: $F(x) = \int_{\text{low}}^{x} f(t)dt$ on the support; $F = 0$ below, $F = 1$ above (write it piecewise).
- **median** m: solve $F(m) = 0.5$. **p-th percentile**: $F(x_p) = p$.
- "at most / below x" → $F(x)$; "more than" → $1 - F(x)$; "between" → $F(b) - F(a)$.

Integrals: $\int x^n dx = \frac{x^{n+1}}{n+1}$, $\int e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x}$.

Practice template $f(x) = kx$, $0 \le x \le 2$: $\int_0^2 kx\,dx = 2k = 1 \Rightarrow k = \frac{1}{2}$. $F(x) = \frac{x^2}{4}$. $P(X < 1) = \frac{1}{4}$. $E[X] = \int_0^2 \frac{x^2}{2} dx = \frac{4}{3}$, $E[X^2] = \int_0^2 \frac{x^3}{2} dx = 2$, $\mathrm{Var} = 2 - \frac{16}{9} = \frac{2}{9}$. Median: $\frac{m^2}{4} = .5 \Rightarrow m = \sqrt{2} \approx 1.414$.

2 Uniform $X \sim U(a, b)$

If you see: "equally likely anywhere between a and b", flat density.

$$f(x) = \frac{1}{b-a},\ a \le x \le b \qquad F(x) = \frac{x-a}{b-a}$$

$$P(c < X < d) = \frac{d-c}{b-a} \qquad E[X] = \frac{a+b}{2} \qquad \mathrm{Var} = \frac{(b-a)^2}{12}$$

where a = smallest possible value · b = largest · c, d = interval asked about · $b - a$ = total width · p = percentile fraction (.90 for 90th).

SOCCER drill starts $U(0, 20)$ min: $P(5 < X < 10) = \frac{5}{20} = .25$; mean 10; percentile: $x_p = a + p(b - a)$.

3 Exponential $X \sim \mathrm{Exp}(\lambda)$

If you see: waiting time until the *first/next* event; λ = **rate per time unit**.

$$f(x) = \lambda e^{-\lambda x},\ x \ge 0 \qquad F(x) = 1 - e^{-\lambda x} \qquad P(X > x) = e^{-\lambda x}$$

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b} \qquad E[X] = \sigma = \frac{1}{\lambda},\ \mathrm{Var} = \frac{1}{\lambda^2}$$

$$\text{percentile: } x_p = \frac{-\ln(1-p)}{\lambda} \qquad \text{median} = \frac{\ln 2}{\lambda}$$

$$\text{memoryless: } P(X > s + t \mid X > s) = P(X > t)$$

where λ = average # of events per time unit · x = waiting time (same units as λ!) · $e \approx 2.718$ · $\ln$ = natural log · p = percentile fraction · s, t = time already waited / extra time.

LECTURE $\lambda = 5$/hour, first customer between 15 and 60 min. **Convert units first**: 0.25 h, 1 h.
$P(0.25 < X < 1) = e^{-5(0.25)} - e^{-5(1)} = .2865 - .0067 \approx$ **.2798**
Rate 5/hour ≠ wait of 5 hours. 2 goals/hour → mean wait $\frac{1}{2}$ h = 30 min.

4 Normal $X \sim N(\mu, \sigma^2)$

If you see: bell-shaped / Normal, μ and SD given. **Trap:** $N(\mu, \sigma^2)$ — 2nd number is the **variance**; $\sigma = \sqrt{\sigma^2}$.

$$\text{standardize: } z = \frac{x - \mu}{\sigma} \qquad \text{un-standardize: } x = \mu + z\sigma$$

where x = raw value (e.g. 0.46 s) · μ = population mean · σ = population SD · z = # of SDs above (+) / below (−) the mean.

Probability: $x \to z \to$ table area (§5). **Find a value/percentile:** area → find it *inside* the table → z → $x = \mu + z\sigma$ (top 10% → left area .90 → $z = 1.28$).

68–95–99.7: within $1\sigma, 2\sigma, 3\sigma$. SOCCER reaction $N(.40, .04^2)$: 68% in .36–.44 s; 95% in .32–.48 s (approx. only — use the table for exact).

5 z-table: area to the LEFT of z

$$\text{table entry} = \Phi(z) = P(Z \le z)$$

where Z = standard Normal (mean 0, SD 1) · $\Phi(z)$ = area LEFT of z.

Row = z to 1 decimal, column = 2nd decimal. $z = 1.96$: row 1.9, col .06 → .9750. Negative z → negative half of the table.

Want	Do	e.g. $z = 1.20$ ($\Phi = .8849$)
left $P(Z < z)$	table	.8849
right $P(Z > z)$	$1 - \Phi(z)$	.1151
between a, b	$\Phi(b) - \Phi(a)$	bigger − smaller
outside $\pm\lvert z\rvert$	$2[1 - \Phi(\lvert z\rvert)]$	$z = 2$: $2(.0228) = .0456$
z from an area	find area inside table	.8849 → 1.20

z^* **(two-sided, look up left area $1 - \frac{\alpha}{2}$):** 80% 1.282 · 90% 1.645 · **95% 1.960** · 98% 2.326 · 99% 2.576. **One-sided** (α in one tail): .10 → 1.282 · .05 → 1.645 · .01 → 2.326.

6 t-table: critical t* for an UPPER tail

Rows = **df**, columns = upper-tail probability $P(T > t^*)$. It does **not** give cumulative areas. Means use t because σ is unknown (we use s) → heavier tails; large df → $t \approx z$.

Pick the column: two-sided CI at C → column $\alpha/2$ (95% → .025; 90% → .05; 99% → .005). One-sided bound/test → column α.
Pick the row: one mean / paired: $n - 1$. Two means: Welch ν, **round DOWN** (e.g. 22.38 → 22). Row missing (e.g. 35 when the table jumps 30→40)? Use the **lower** df row (conservative).

p-value by bracketing: in your df row find the two critical values around $|t|$ → p is between their column headings (one-tail). **Double both** for H_a :≠. Bigger than every entry → p < smallest heading.
e.g. $df = 35$: $2.030 < \mathbf{2.40} < 2.438$ → $.01 < p < .025$ (one-tail).

Inference: CIs · Tests · Worked Examples · MC

8 CI master rule

$$\text{CI} = \text{PE} \pm \underbrace{c^* \times SE}_{\text{margin of error } E}$$

1. Parameter? → PE + SE formula (table A, side 1).
2. Critical value: proportions z^* (§5), means t^* (§6).
3. $E = c^* \cdot SE$; interval $(\text{PE} - E,\ \text{PE} + E)$.
4. Conditions (§13) + interpretation sentence.

Interpretation template: "We are 95% confident that the true *[parameter in context]* is between ___ and ___."
Never: "95% probability the parameter is in this interval." Long-run meaning: 95% of intervals built this way capture the parameter.

Reading a given CI (L, U): $\text{PE} = \frac{L+U}{2}$, $E = \frac{U-L}{2}$, $SE = E/c^*$, width $= 2E$.
Width ↑ when confidence ↑ or n ↓ or variability ↑. Width $\propto 1/\sqrt{n}$ (×4 n → half width).

One-sided bounds ($\alpha = 1 - C$ in ONE tail)

lower: $(\text{PE} - c^*SE,\ \infty)$ upper: $(-\infty,\ \text{PE} + c^*SE)$

c^*: z with left area C (90% → 1.282), or t from column α (90% → .10 column). Professor's equivalent: negative z with left area α, compute PE + z·SE.
All five CI formulas are in table A (side 1). t^*_{n-1}: subscript = df, not "t* minus 1". Paired: n = # pairs.

Difference CIs: look at 0
Contains 0 → "no difference" plausible · all > 0 → group 1 larger · all < 0 → group 1 smaller.

10 Sample size for a proportion

$$E = z^*\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \implies n = \left\lceil \frac{(z^*)^2\,\hat{p}(1-\hat{p})}{E^2} \right\rceil$$

Use a prior $\hat{p}$ if given, else .5 ($.5 \times .5 = .25$ is the max → safest). **Round UP**, even 9603.1 → 9604.

E ($\hat{p} = .5$)	.01	.02	.03	.04	.05
90% (1.645)	6766	1692	752	423	271
95% (1.96)	**9604**	2401	1068	601	385
99% (2.576)	16590	4148	1844	1037	664

Halve E → 4× the sample. $\frac{1.96^2(.5)(.5)}{.01^2} = 9604$.

+ Reverse z lookups (percentiles & cutoffs)

Want	left area to find	z
top 10% / 90th pct	.9000	1.28
top 5% / 95th pct	.9500	1.645
top 2.5%	.9750	1.96
top 1%	.9900	2.33
bottom 10% / 5%	.1000 / .0500	−1.28 / −1.645
middle 90% / 95% / 99%	tails .05 / .025 / .005	±1.645 / ±1.96 / ±2.576
quartiles Q1 / Q3	.2500 / .7500	−0.67 / 0.67

Then $x = \mu + z\sigma$ (individual) or $\bar{y} = \mu + z\,\sigma/\sqrt{n}$ (sample mean). Area between two closest table entries → pick the closer one (or average).

7 Sampling distributions, SE & Welch df

$$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right),\ SE(\bar{y}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right),\ SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

where $\bar{Y}$ = sample mean (random) · c = cutoff in the question · other symbols: key on side 1.
SD = spread of individuals; **SE** = spread of a statistic over repeated samples. n ×4 → SE ÷2.

CLT probability: $P(\bar{Y} > c) = 1 - \Phi\left(\frac{c-\mu}{\sigma/\sqrt{n}}\right)$. Practice: $\mu = .40, \sigma = .04, n = 16$: SE $= .01$; $P(\bar{Y} > .42) = 1 - \Phi(2.00) = .0228$.

Sample SD (e.g. of paired d_i): $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}\ \text{("crazy df")}$$

where s_1, s_2 = sample SDs · n_1, n_2 = group sizes · x_i = each data value · $\bar{x}$ = their mean · ν ("nu") = Welch df.
Compute $A = s_1^2/n_1$, $B = s_2^2/n_2$ first: $\nu = (A+B)^2/\left(\frac{A^2}{n_1-1} + \frac{B^2}{n_2-1}\right)$. If allowed, conservative shortcut $df = \min(n_1, n_2) - 1$.

11 Hypothesis test workflow

1. H_0 (has "=", the claim/default) and H_a (<, >, ≠) — choose before seeing data.
2. Check conditions (§13).
3. Test statistic = $\frac{\text{estimate} - \text{null value}}{SE \text{ under } H_0}$ (table A).
4. Tail from H_a → p-value from the right table.
5. $p \le \alpha$ → **reject** H_0; $p > \alpha$ → fail to reject H_0.
6. Conclusion in context.

Wording → H_a: "less / decreased / lower" → < · "more / increased / slower(time ↑)" → > · "different / changed / not equal" → ≠.
Conclusion template: "Since p = ___ (< or >) α = ___, we (reject / fail to reject) H_0. There is (enough / not enough) evidence that *[H_a in context]*."

12 p-value: tail × table

H_a	**z-test (table = left area)**	**t-test (bracket in df row)**
< left	$p = \Phi(z)$	one-tail area beyond $\lvert t\rvert$
> right	$p = 1 - \Phi(z)$	one-tail area beyond $\lvert t\rvert$
≠ two	$p = 2[1 - \Phi(\lvert z\rvert)]$	2 × one-tail bracket

e.g. $z = -1.20$: H_a: <→ .1151; H_a: >→ .8849; H_a: ≠→ .2302.

Rejection regions (critical-value method): $\alpha = .05$ — left: $z < -1.645$ · right: $z > 1.645$ · two: $|z| > 1.96$. For t: same idea with t^* from the df row (column α one-sided, $\alpha/2$ two-sided).
CI ↔ test: two-sided test at α rejects $H_0: \theta = \theta_0$ exactly when θ_0 is outside the $(1-\alpha)$ CI.

13 Conditions (write them!)

	Random	**10%**	**Shape / counts**	**Indep. groups**
1 prop	✓	✓	CI $n\hat{p}, n\hat{q} \ge 10$; test $np_0, nq_0 \ge 10$	—
2 props	both	both	S/F for both	✓
1 mean	✓	✓	Normal-ish; skew → $n > 30$; heavy skew → $n > 100$	—
2 means	both	both	same, both groups	✓
paired	✓	✓	differences Normal-ish	pairs dependent!

10%: $n \le 0.10N$ (e.g. $n = 36$ needs population ≥ 360).

14 Worked examples

LECTURE One proportion test (voters): $H_0: p = .52$, $H_a: p < .52$, $n = 10{,}000$, $\hat{p} = .514$.

$$z = \frac{.514 - .52}{\sqrt{.52(.48)/10000}} = \frac{-.006}{.004996} \approx -1.20$$

left tail: $p = \Phi(-1.20) = .1151$

$.1151 > .05$ → fail to reject; not enough evidence support is below 52%. CI version (uses $\hat{p}$): $.514 \pm 1.96(.004998)$ → (50.42%, 52.38%).

LECTURE Two proportions (cats): F 13/45 = .2889, M 16/74 = .2162, $H_a: p_F - p_M > 0$, unpooled SE:

$$z = \frac{.2889 - .2162}{\sqrt{\frac{.2889(.7111)}{45} + \frac{.2162(.7838)}{74}}} = \frac{.0727}{.0828} \approx 0.878$$

right tail: $p = 1 - \Phi(0.88) = 1 - .8106 = .1894 > .10$ → fail to reject (insufficient evidence).

LECTURE One mean (HS GPA): $H_0: \mu = 4.1$, $H_a: \mu \ne 4.1$, $n = 114$, $\bar{y} = 4.2317$, $s = .3664$:

$$t = \frac{4.2317 - 4.1}{.3664/\sqrt{114}} = \frac{.1317}{.03432} \approx 3.84, \quad df = 113$$

Row 113 missing → use lower row (e.g. 100). t beyond the last column (one-tail .001) → two-sided $p < .002$ (lecture: $p < .001$; exact .0002). $\alpha = .02$ → reject: mean differs from 4.1.

LECTURE Two means (1st-gen vs not): $n_1 = 17, \bar{y}_1 = 4.1467, s_1 = .3495$; $n_2 = 102, \bar{y}_2 = 4.2466, s_2 = .369$; $H_a: \mu_1 - \mu_2 < 0$.

$$t = \frac{4.1467 - 4.2466}{\sqrt{\frac{.3495^2}{17} + \frac{.369^2}{102}}} = \frac{-.0999}{.0923} \approx -1.083$$

$\nu \approx 22.38 \to 22$, left tail: $p \approx .145$

$.145 > .03$ → fail to reject.

SOCCER One-mean t-test: claim $\mu = .400$ s, suspect slower: $H_a: \mu > .400$; $n = 36$, $\bar{y} = .424$, $s = .060$, $\alpha = .05$.

$$t = \frac{.424 - .400}{.060/\sqrt{36}} = \frac{.024}{.010} = 2.40,\ df = 35$$

Row 35: $2.030 < 2.40 < 2.438$ → $.01 < p < .025 < .05$ → reject; mean reaction time exceeds .400 s. (Row 30 instead: 2.042 < 2.40 < 2.457, same result.)

SOCCER One-proportion CI: 62 of 100 prefer morning training, 95%:

$$SE = \sqrt{\frac{.62(.38)}{100}} = .04854, \quad E = 1.96(.04854) = .0951$$

$$.62 \pm .0951 = (.5249,\ .7151)$$

S/F: 62, 38 ≥ 10 ✓. Testing $H_0: p = .50$ instead → SE $= \sqrt{.5(.5)/100} = .05$, $z = \frac{.62-.50}{.05} = 2.40$.

SOCCER Paired recipe: each keeper's d_i = after − before → $\bar{d}$, s_d (§7 formula) → $t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}$, df = pairs − 1.
Never use the two-sample formula on paired data.

15 MC must-knows

- **Parameter** (p, μ; population, unknown) vs **statistic** ($\hat{p}, \bar{y}, s$; sample).
- **p-value** = P(data this extreme or more | H_0 true). It is NOT P(H_0 true).
- Fail to reject ≠ "accept H_0" / ≠ proof H_0 is true.
- **Type I**: reject a true H_0 (false alarm), prob $= \alpha$. **Type II**: fail to reject a false H_0 (miss), prob $= \beta$. **Power** $= 1 - \beta$ = P(correctly reject false H_0).
- Power ↑ with larger n, larger effect size, larger α (but Type I risk ↑). ↓α → ↑β.
- Confidence level ↑ → wider CI; n ↑ → narrower CI.
- t vs z: t heavier tails, depends on df; df → ∞ gives z.
- CI uses $\hat{p}$ in SE; test uses p_0 in SE.
- Continuous: $P(X = x) = 0$; probability = area; density can exceed 1 (area can't).
- Uniform flat · Exponential = time until first event (constant rate) · Normal symmetric bell.
- H_a is chosen before seeing data; tail comes only from H_a.

16 Trap list

- p (proportion) ≠ p-value · Var vs SD in $N(\mu, \sigma^2)$ · minutes vs hours for λ.
- z-table is LEFT area; t-table is UPPER-tail critical values — don't read t like z.
- Two-sided: split α (.025 each tail) for CIs; double one-tail p for ≠ tests.
- Paired ≠ two independent groups. Means → t, proportions → z.
- Sample size: round **up**; E as decimal.
- Zero in a difference CI = no difference plausible.